

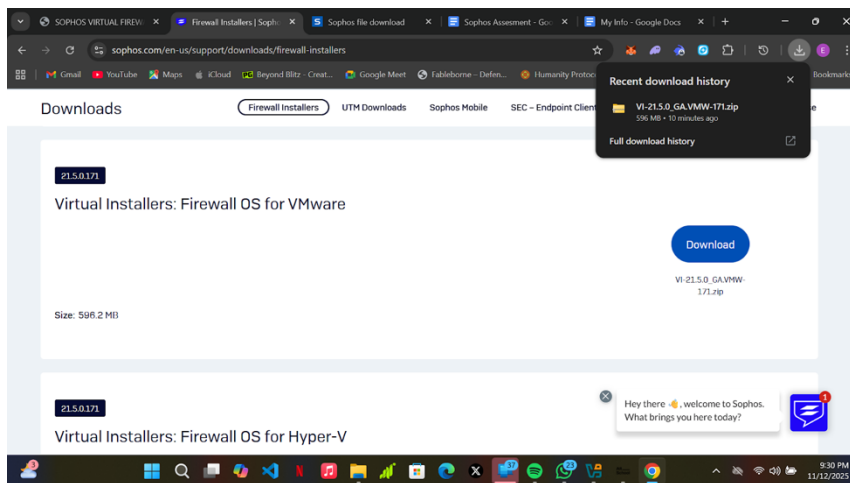
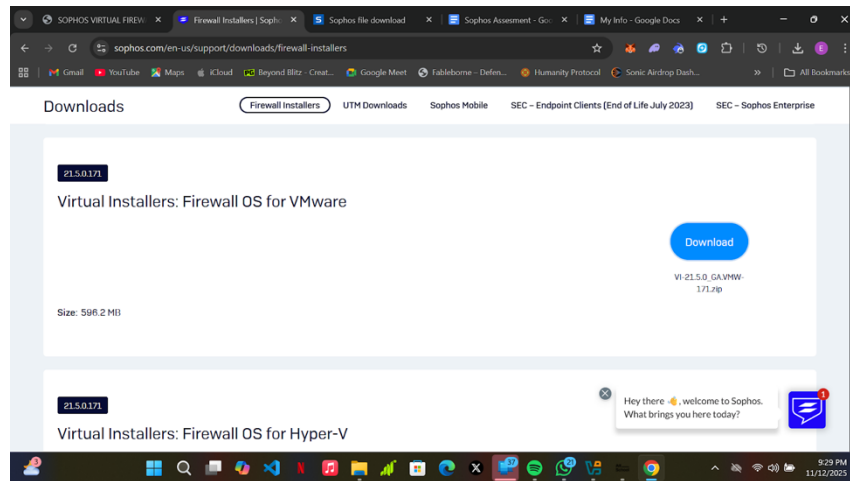
MAC-DONALD ONYEWUCHI EGBUNA  
SOPHOS NEXT-GEN FIREWALL  
INSTALLATION, CONFIGURATION & SETUP

ALT/SOE/025/4248

# SOPHOS FIREWALL DOWNLOAD AND PRE INSTALLATION

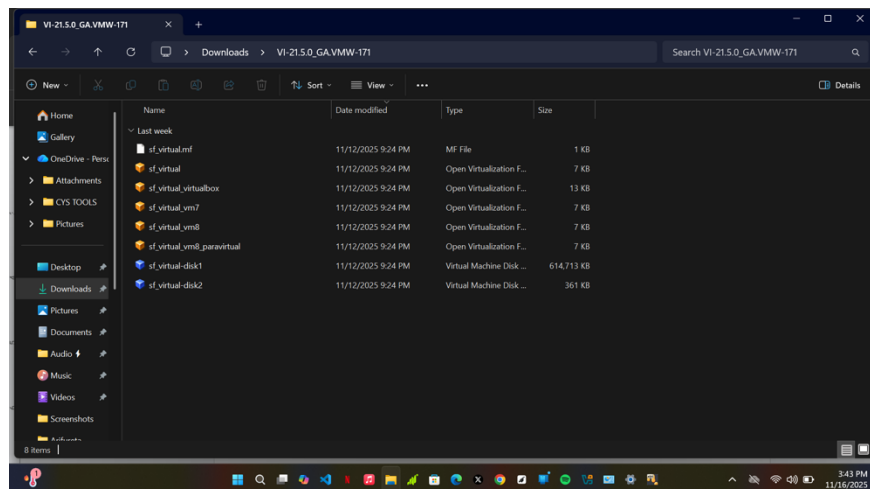
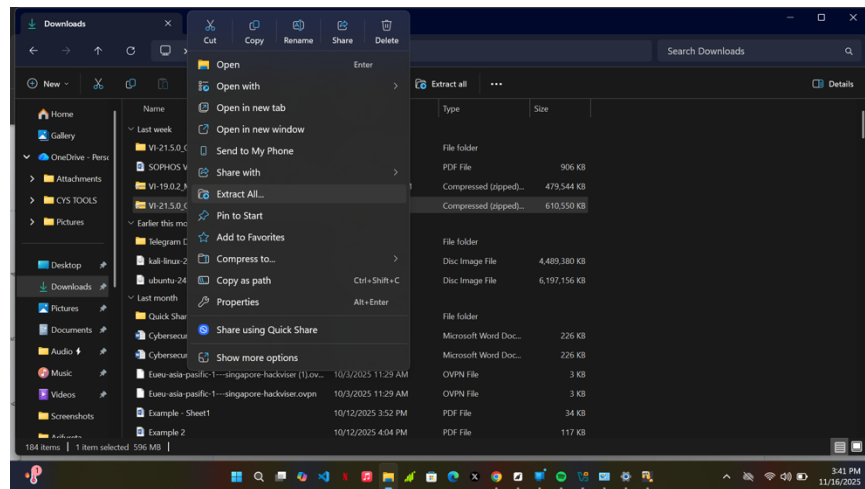
## Step 1 : Firewall Installers

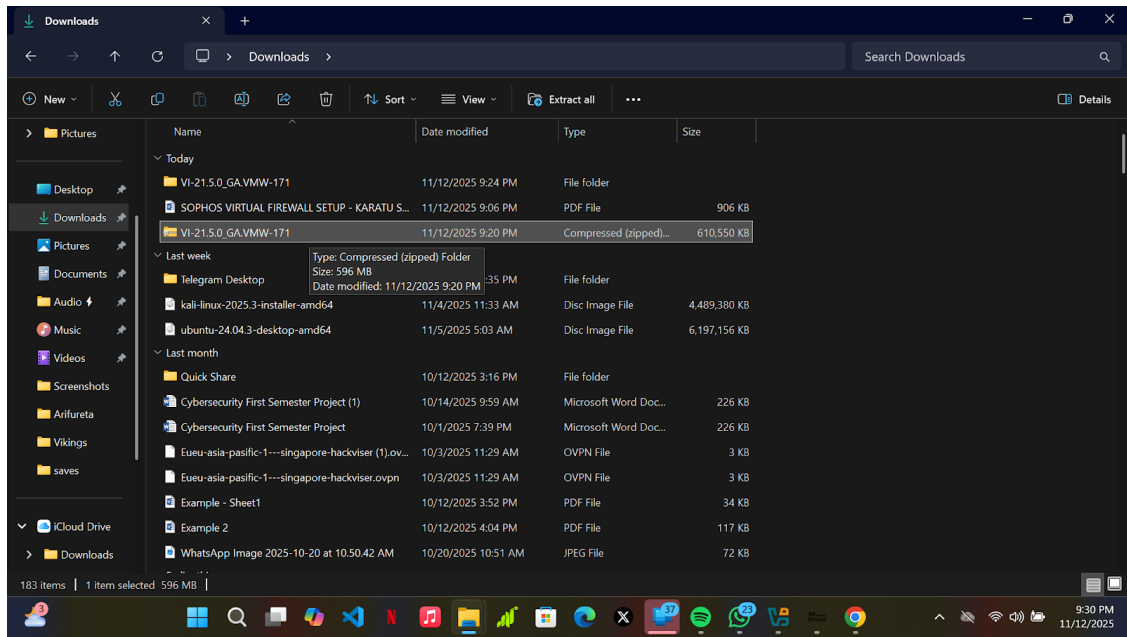
- I. Launch a browser (Firefox,chrome or any browser of choice)
- II. Visit the Sophos Firewall Installers webpage using the link:  
<https://www.sophos.com/en-us/support/downloads/firewall-installers>
- III. Scroll down and locate the card titled “Virtual Installers: Firewall OS for VMware” and click the contained Download button. This opens the End User.



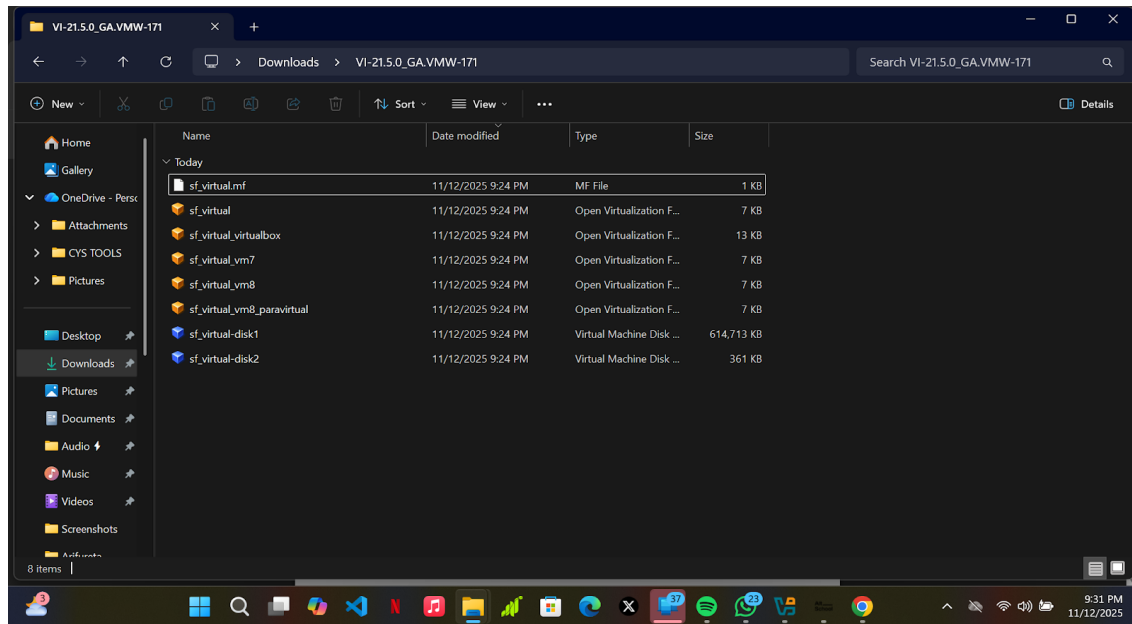
## Step 2: Pre-Installation Setup

- I. Launched the ZIP file, VI-21.5.0\_GA.VMW-171.zip, using a file decompressor/extractor software.
- II. Extracted the contents of the ZIP file to my desired location







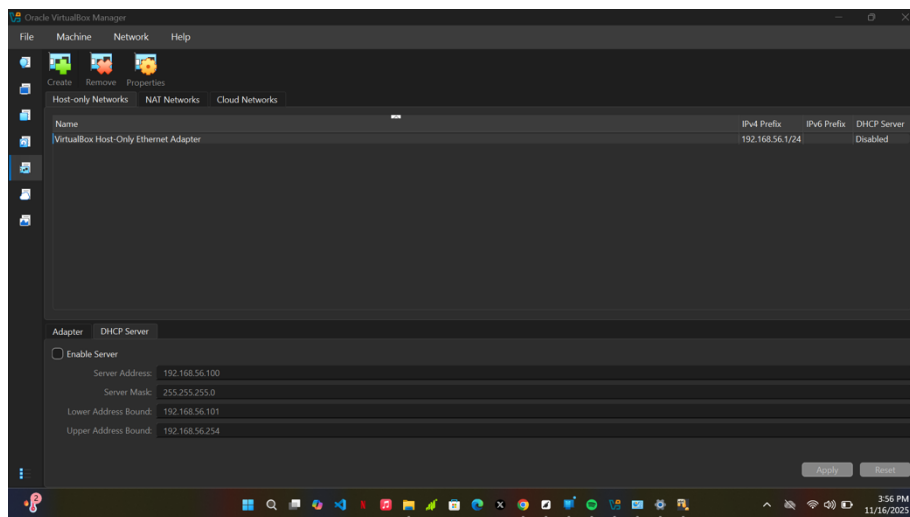
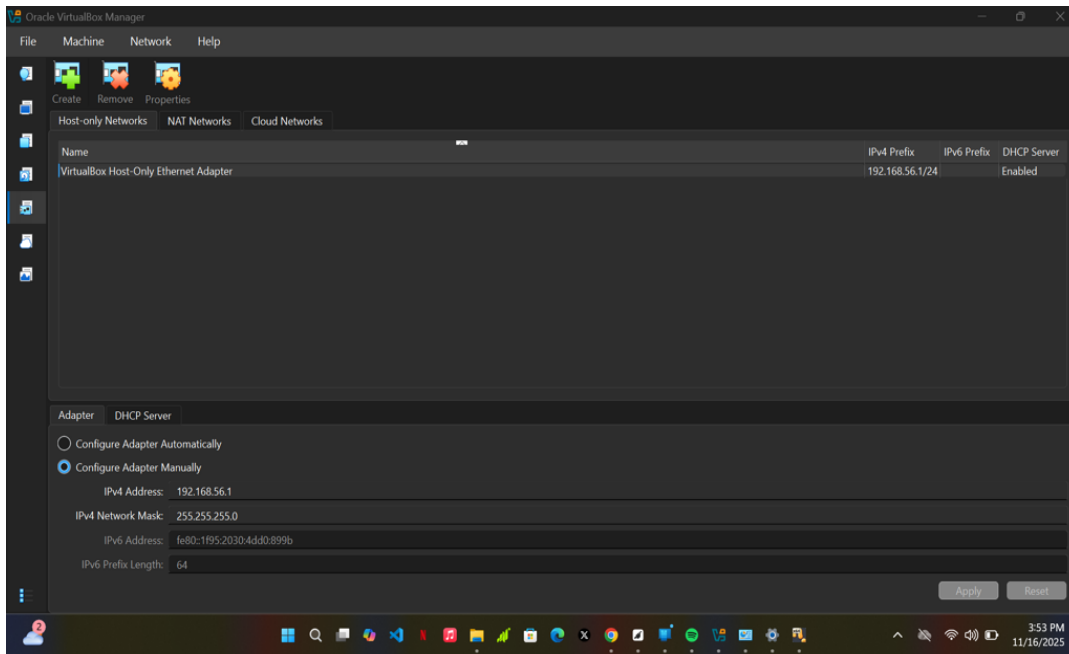


### Step 3: VIRTUAL BOX NETWORK CONFIGURATION

#### Step 3:

- Imported the previously downloaded firewall VM
- Open your virtualbox and select “file” at the top-left corner and click on “import appliances”. This is what you will see
- Click on the file icon
- Navigate to where you extracted the sophos’ firewall virtual machine and select sf\_virtual\_virtualbox.ovf .
- Click on open.
- After clicking on open, click on finish to import the virtual machine in your virtualbox.
- The image below is what you should see after the machine has been imported. *Kindly note that it appears as “Virtual\_SF 1” on my virtualbox because I have installed it before. Ideally, it should appear as “Virtual\_SF” on your virtualbox*
- Before launching the virtual machine, navigate to File >> Tools >> Network manager.

- Click on Network manager, then click on Host-only Networks, then click on Adapter , select “configure Adapter manually” and insert the IP address and subnet mask verbatim (use the same values)



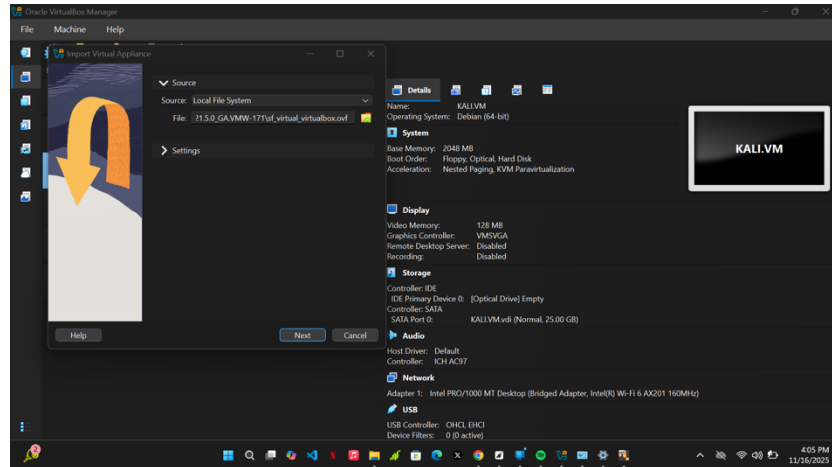
- I. Unchecked the DHCP option
- II. Configured my IPV4 address to 192.168.56.1
- III. Configured the subnet mask to 255.255.255.0



# FIREWALL MACHINE CREATION AND NETWORK CONFIGURATION

## Step 5:

- Imported the previously downloaded firewall VM



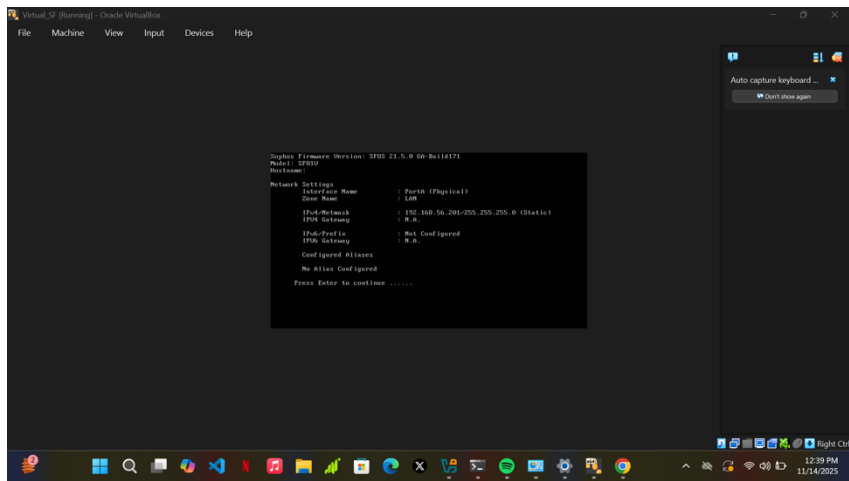
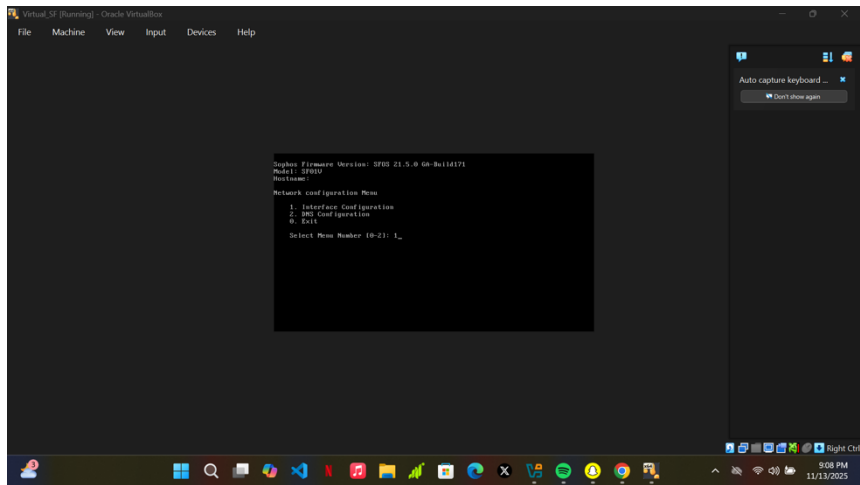
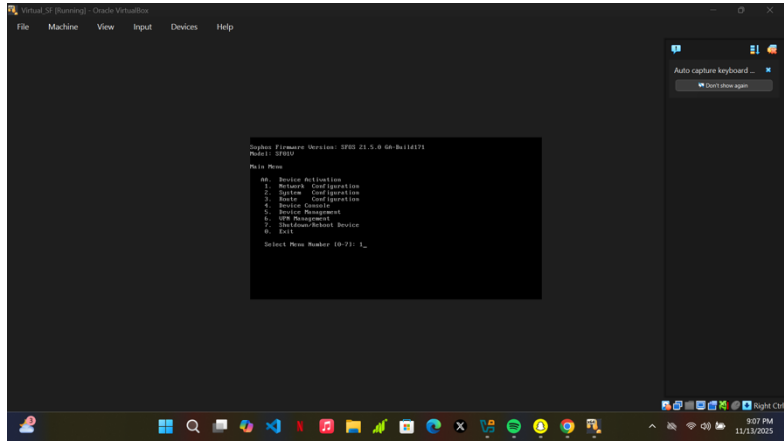
## Step 6:

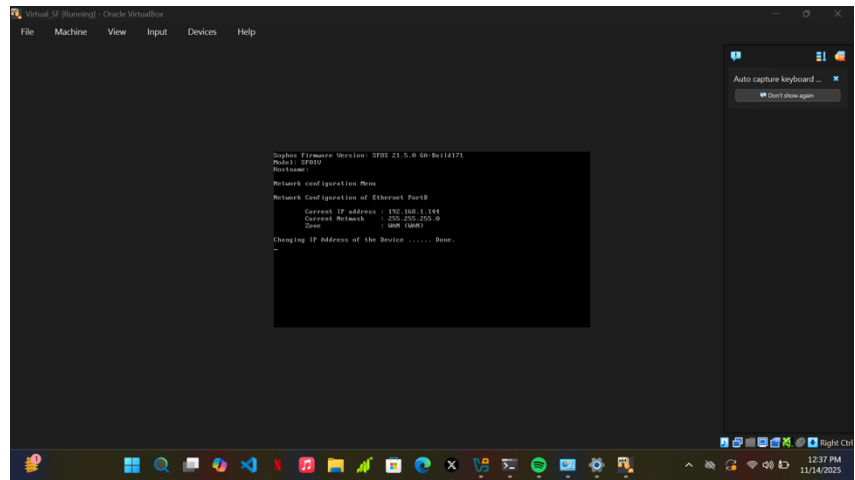
### Firewall Network configuration

- I. Configured my Adapter1, making to apply the “HOST-ONLY ADAPTER” setting
- II. Configured my Adapter 2, making sure to apply “BRIDGED ADAPTER” setting

## FIREWALL OPERATING SYSTEM SETUP

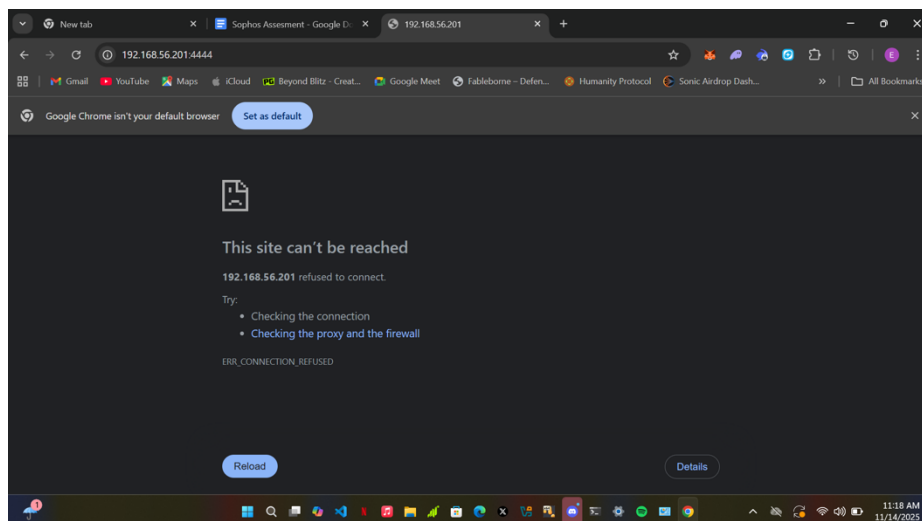
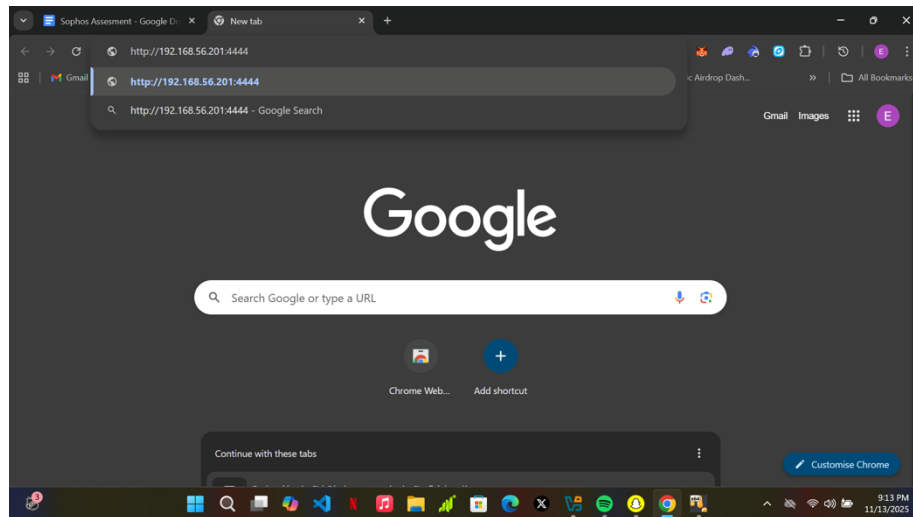
- I. Launched and installed the Sophos firewall operating system
- II. Set the PORT A IP to 192.168.56.201 and the Subnet mask to 255.255.255.0
- III. Left the PORT B IP and subnet mask to the default IP and subnet mask.



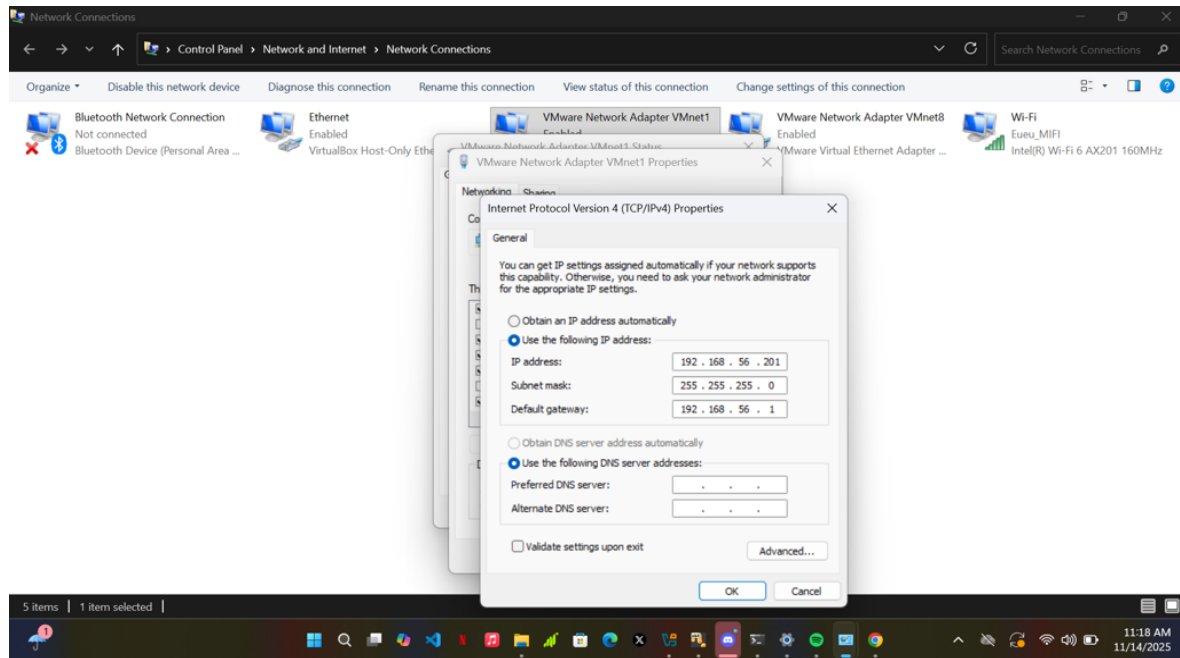


## SETTING UP FIREWALL IN WEB ADMIN CONSOLE

At first, after deploying and configuring the sophos firewall, the webpage was not reachable.

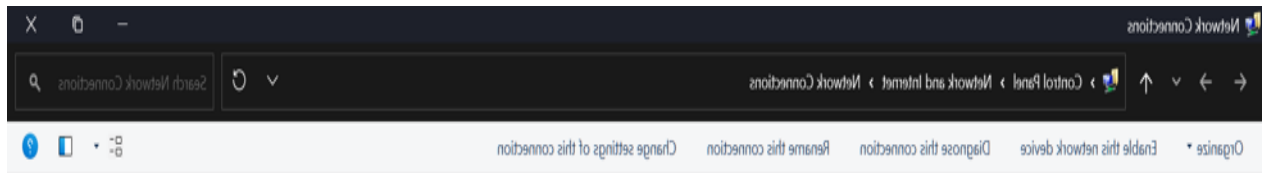


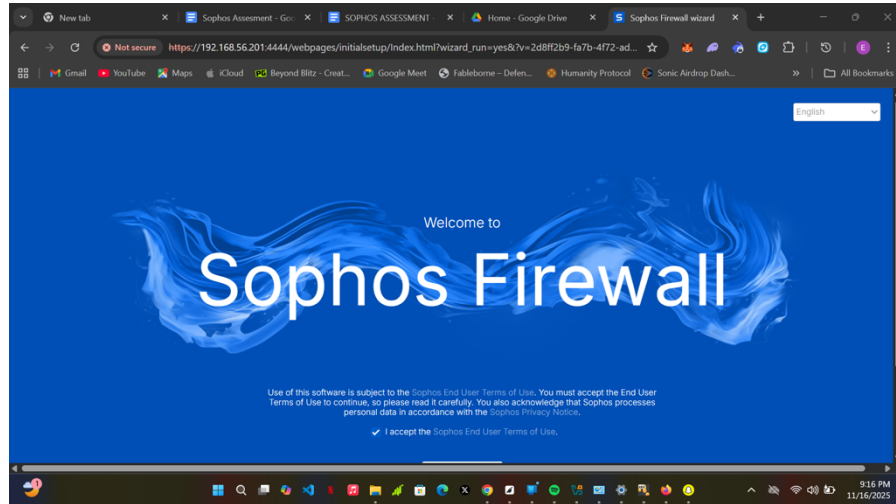
I tried to troubleshoot by making sure the VMnet1 on my host PC had the same IP address as that of the virtual machines, but it had a different IP altogether and I had to reconfigure the IP.



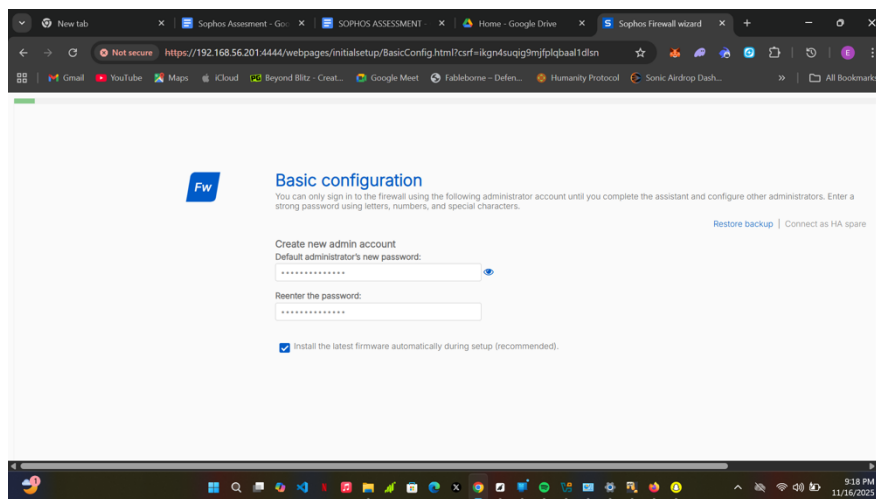
Even after reconfiguring the VMnet1 IP, the web page was still not accessible. Checked the network connections again and realized that the network adapters were conflicting (VMware and VirtualBox adapter). To rectify this issue, i disabled **“VMWARE NETWORK ADAPTER VMnet1”** and was able to access the web page of the sophos firewall IP address.







## Basic Configuration setup



## Create a secure storage master key

The screenshot shows a web browser window with the URL `https://192.168.56.201:4444/webpages/initialsetup/SSMKConfig.html?csrf=qcnd00b9so2888mlv1kjmog115`. The page title is "Secure storage master key" and it features a blue "FW" logo. The text explains that the master key provides extra protection for account and password details. There are two input fields: "Create secure storage master key:" and "Repeat the master key:". Below these fields, a warning icon and text state: "You can't recover a lost master key or restore backups and import configurations created with it. For more details, go to [Secure storage master key](#)." A checkbox labeled "I have stored the master key in a password manager or another secure location" is checked. The Windows taskbar at the bottom shows the time as 11:41 PM on 11/16/2025.

**Secure storage master key**

The master key provides extra protection for the account and password details stored in the firewall. You need the master key to restore backup and import configurations.

Create secure storage master key:

Repeat the master key:

**Warning:** You can't recover a lost master key or restore backups and import configurations created with it. For more details, go to [Secure storage master key](#).

☒ I have stored the master key in a password manager or another secure location

## Name and Timezone

The screenshot shows the "Name and time zone" screen of the Sophos Firewall wizard. The URL is `https://192.168.56.201:4444/webpages/initialsetup/NameTimezone.html?csrf=qcnd00b9so2888mlv1kjmog115`. It features a blue "FW" logo and instructions to enter a firewall name and select a time zone. The "Firewall name" field contains "MacDonnell-Firewall". The "Time zone" section includes a world map with a red pin in Africa and a dropdown menu showing "Africa/Lagos". Below the map, the "Current time" is displayed as "Sunday, November 16, 2026, 11:48 PM". The Windows taskbar at the bottom shows the time as 11:50 PM on 11/16/2025.

**Name and time zone**

Enter a firewall name. We recommend that you use a fully qualified domain name (FQDN) that points to this device.

Firewall name

MacDonnell-Firewall

Time zone

You can choose the time zone on the map, or from the dropdown list below. It is important to choose the correct time zone. It affects the scheduled events, logs, and reports.

Africa/Lagos

Current time: Sunday, November 16, 2026, 11:48 PM

Acquire Serial number for free trial access

The screenshot shows the Sophos Firewall free trial registration page. The page has a blue header with the Sophos Firewall logo and navigation links: Overview, Features, Compare Models, Ecosystem, Central, Free Trial, and How to Buy. The main content area is titled "Sophos Firewall free trial" and includes a sub-header "Sophos Firewall spots hidden risks other firewalls miss and automatically isolates active threats". Below this, there are three bullet points describing the features: "Blocks unknown threats with a comprehensive suite of advanced protection that includes intrusion prevention system (IPS), active threat prevention (ATP), sandboxing, dual antivirus (AV), web and app control, anti-phishing, a full-featured web application firewall, and more", "Automatically responds to incidents by instantly identifying infected systems and isolating them until they can be cleaned up", and "Exposes hidden risks on your network, including unknown apps, top-risk users, advanced threats, suspicious payloads, and more". To the right of the text is a registration form with the following fields: First Name (Mac-Donald), Last Name (E), Business Email (echicheta@gmail.com), Company (Altschool Africa), Number of Employees (0-50), Job role (Home User/Student/F...), Phone number (08034064577), and Country (Nigeria). A chat bubble on the right says "Hey there, welcome to Sophos. What brings you here today?". The Windows taskbar at the bottom shows the time as 11:59 PM on 11/16/2025.

Sophos Firewall

Overview Features Compare Models Ecosystem Central Free Trial How to Buy

## Sophos Firewall free trial

Sophos Firewall spots hidden risks other firewalls miss and automatically isolates active threats

- Blocks unknown threats with a comprehensive suite of advanced protection that includes intrusion prevention system (IPS), active threat prevention (ATP), sandboxing, dual antivirus (AV), web and app control, anti-phishing, a full-featured web application firewall, and more
- Automatically responds to incidents by instantly identifying infected systems and isolating them until they can be cleaned up
- Exposes hidden risks on your network, including unknown apps, top-risk users, advanced threats, suspicious payloads, and more

First Name \* Mac-Donald

Last Name \* E

Business Email \* echicheta@gmail.com

Company \* Altschool Africa

Number of Employees \* 0-50

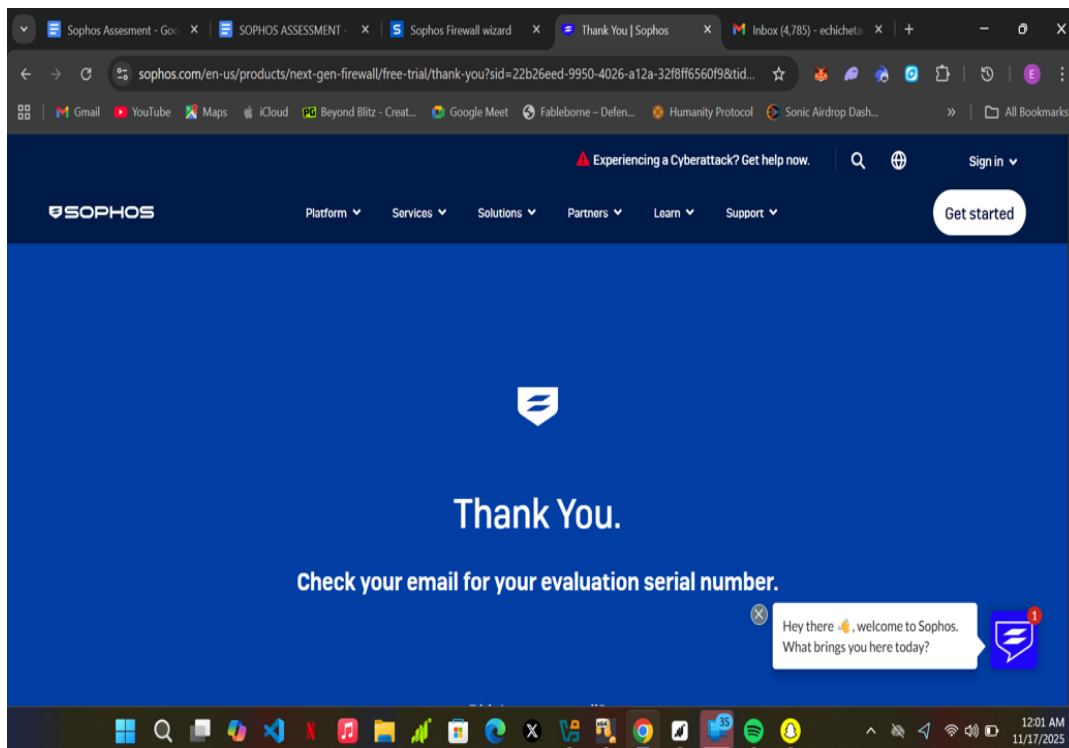
Job role \* Home User/Student/F...

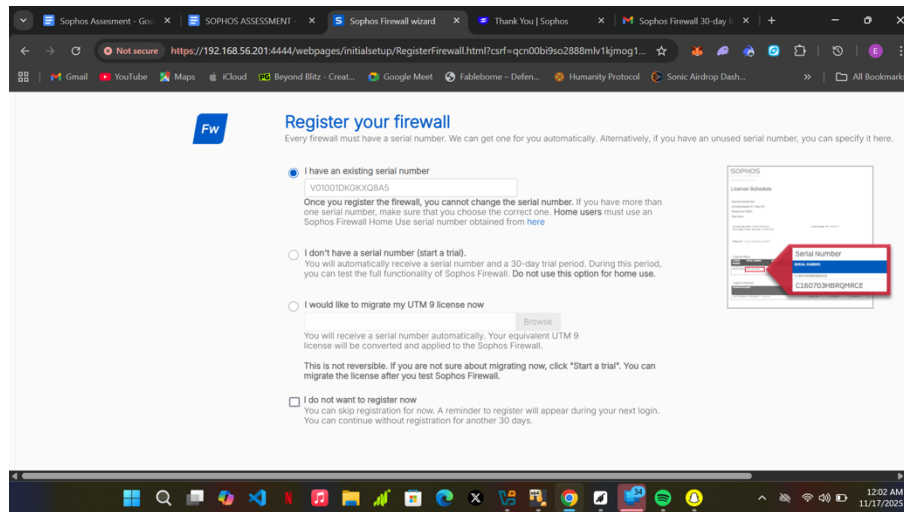
Phone number \* 08034064577

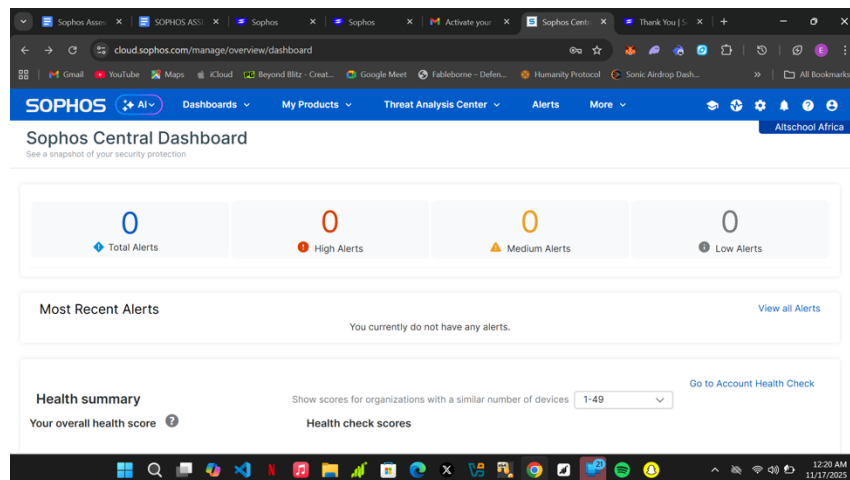
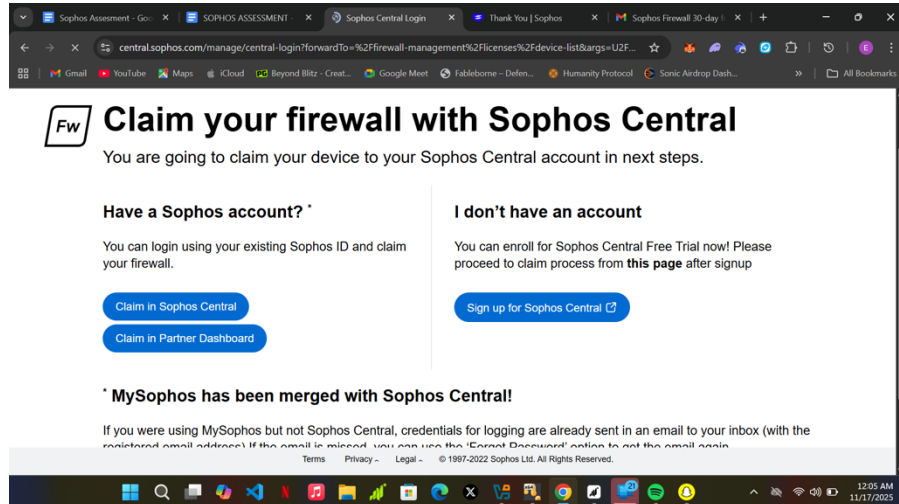
Country \* Nigeria

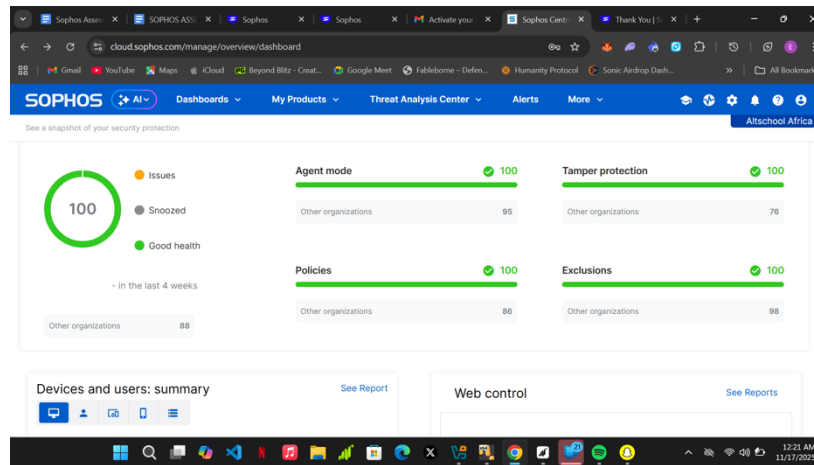
Hey there, welcome to Sophos. What brings you here today?

11:59 PM 11/16/2025









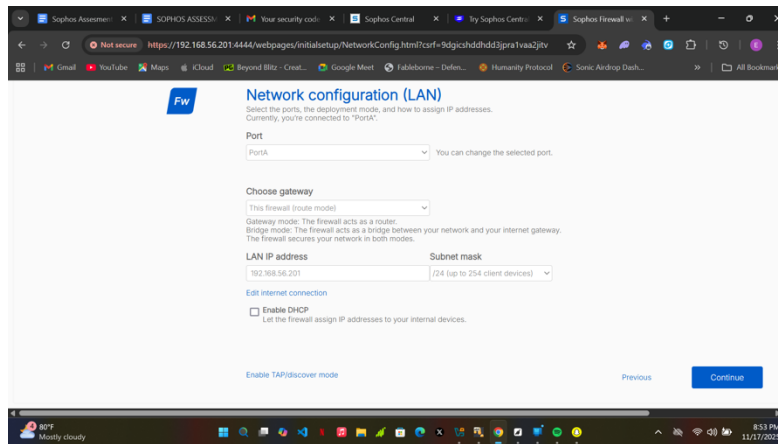
The screenshot shows the 'Basic setup is complete' page in Sophos Central. It includes a 'Serial number' field and a table of 'Licensed subscriptions'. The table lists various security features and their expiration dates.

| Subscription                    | Status     | Expiration date |
|---------------------------------|------------|-----------------|
| <b>Xtream Protection bundle</b> |            |                 |
| Base Firewall                   | Evaluating | Dec 12, 2025    |
| Network Protection              | Evaluating | Dec 12, 2025    |
| Web Protection                  | Evaluating | Dec 12, 2025    |
| NDR Essentials for Firewall     | Evaluating | Dec 12, 2025    |
| Zero-Day Protection             | Evaluating | Dec 12, 2025    |
| Central Orchestration           | Evaluating | Dec 12, 2025    |
| DNS Protection                  | Evaluating | Dec 12, 2025    |
| Enhanced Support                | Evaluating | Dec 12, 2025    |

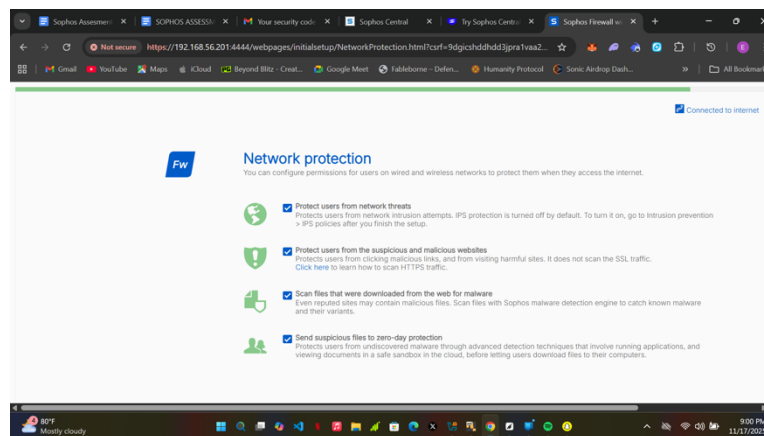
The Windows taskbar at the bottom shows the time as 8:49 PM on 11/17/2025.

## Network Configuration (LAN)

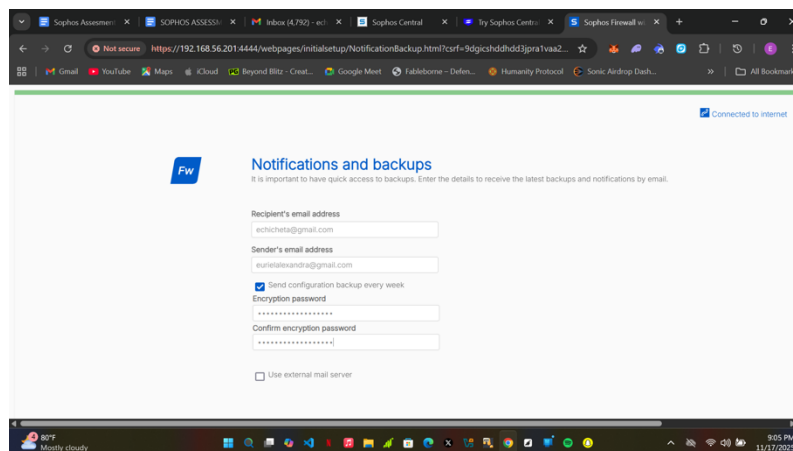
- I. Confirmed that the IP in the LAN IP address text box is the same as the previously configured PORTA IP address.
- II. Unchecked the DHCP check box and continued



Checked all the boxes on the network protection page

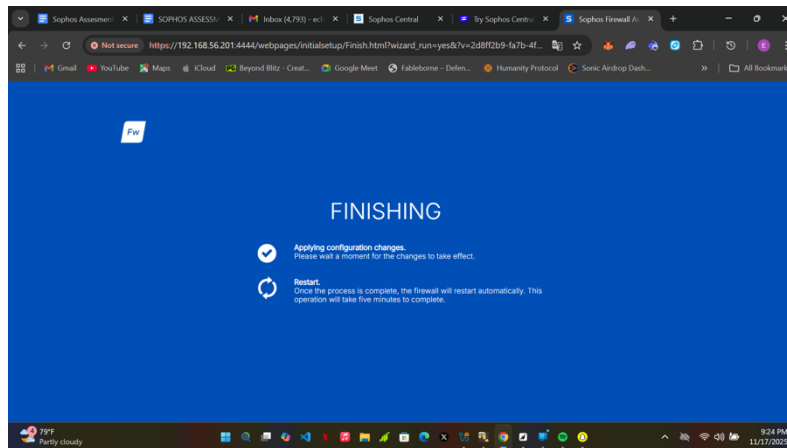
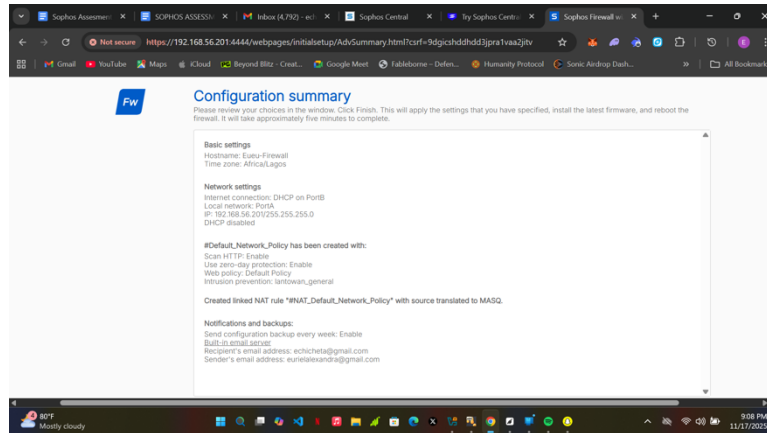


Notifications and Backups page setup

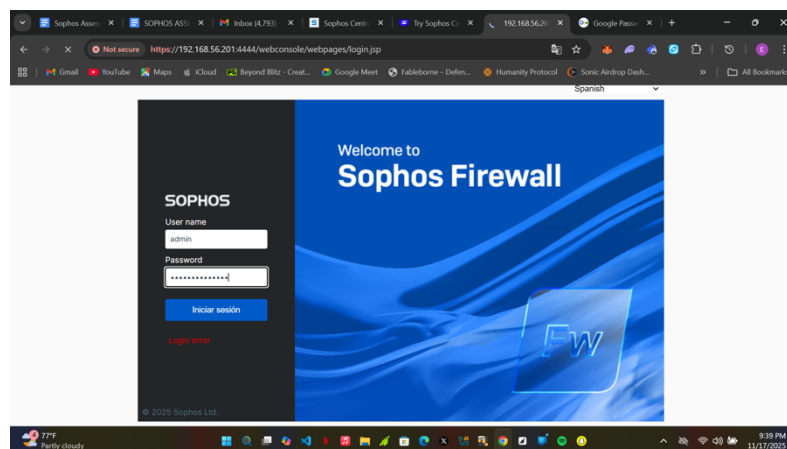




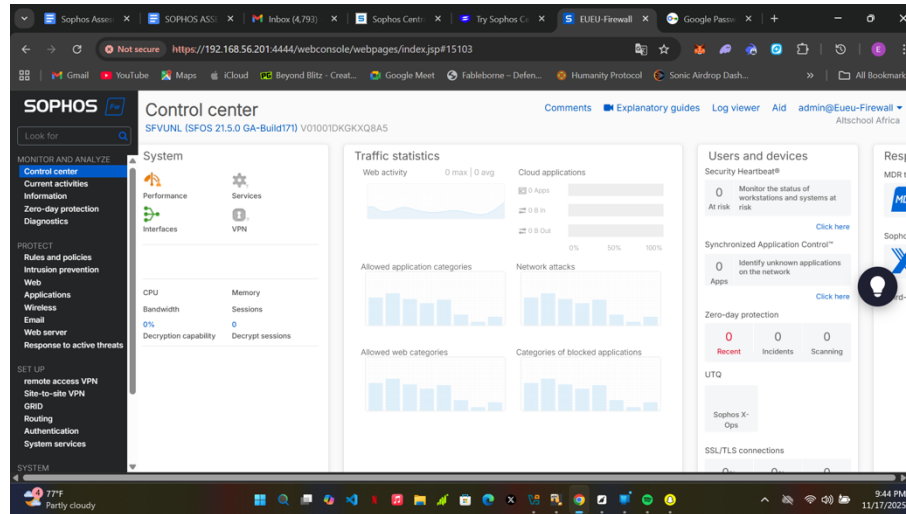
## Configuration summary



## Sign up page



## Sophos firewall control center



## POLICY CONFIGURATION ON SOPHOS FIREWALL

### POLICY 1: Blocking Outbound Web Access for Kali VM (192.168.1.120 )

Step 1:

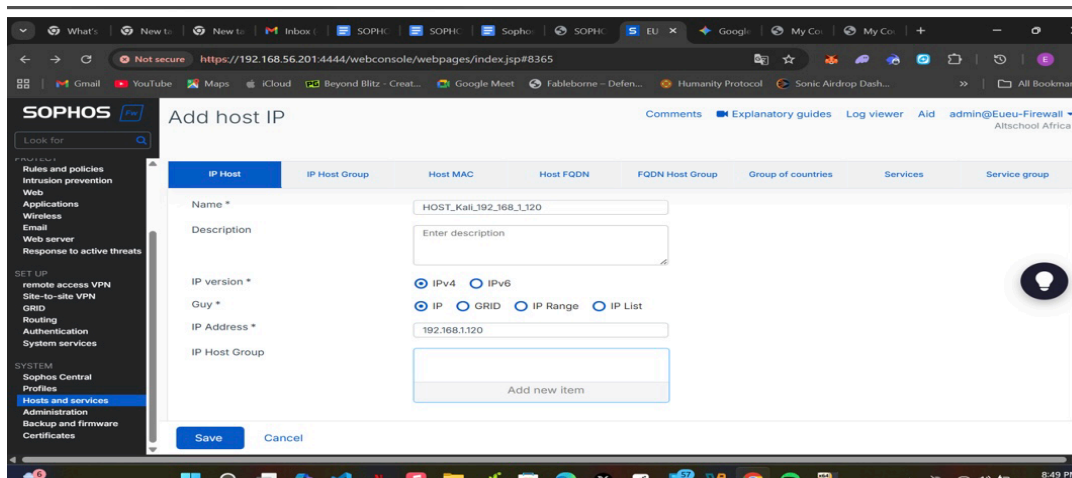
- On the Sophos Web Console, look at the left sidebar menu.
- Click on Host and services
- Select the ip host tab and click the add button

Define the Kali VM's specific IP address (**192.168.1.120**) as a reusable object named **HOST\_Kali\_192\_168\_1\_120**.

- IP version set to **IPv4**
- Type (guy) set to **IP**
- IP address **192.168.1.120**



- Select the ip host tab and click the add button
- Define the Kali VM's specific IP address (**192.168.1.120**) as a reusable object named **HOST\_Kali\_192\_168\_1\_120**.
- IP version set to **IPv4**
- Type (guy) set to IP
- IP address **192.168.1.120**



- Click **Save** at the bottom of the window.

Step 2:

- Navigation Path: Rules and Policies →→Firewall Rules → Add Firewall Rule  
→ New Firewall Rule

### A. General Settings

- Rule name: Should be **DROP\_Kali\_Web\_Traffic**.
- Action: Select Rule out (This is the silent block/drop action).
- Rule position: Select Top
- Log Firewall Traffic: Check this box (you'll need to scroll down).

### B. Origin (Source Section)

- Areas of origin: Select LAN Zone.
- Source devices and networks: Select the object **HOST\_Kali\_192\_168\_1\_120**.

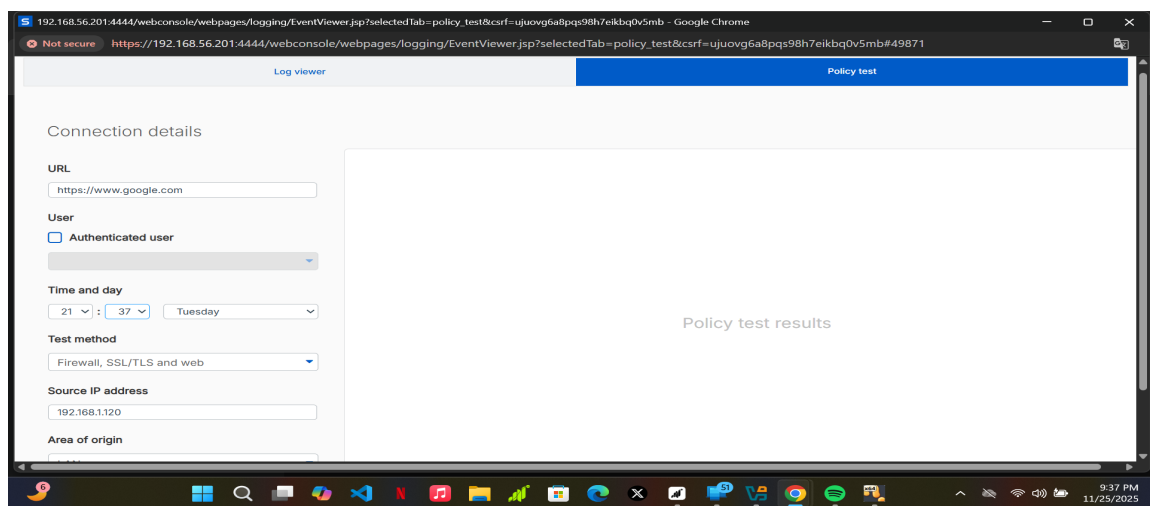
### C. Destination and services

- Destination areas: Click Add new item → Search for and select the WAN Zone.

- Destination networks: Click Add new item → Search for and select the object Any.
- Services: Click Add new item → Search for and select the service group Web Browsing

Click the Save button.

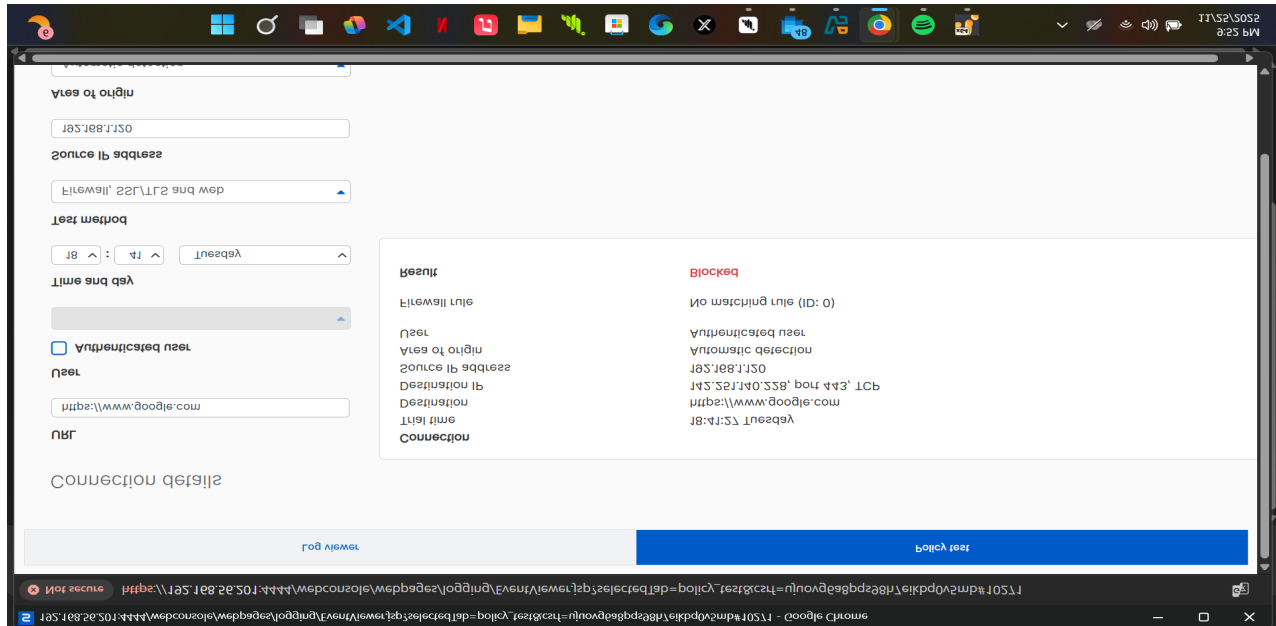
- Finally, Analyze the Policy Test Results and confirm the rule set is working



At first the rule configured on the sophos firewall didn't work



- Action: Find your rule, DROP\_Kali\_Web\_Traffic, and click the Edit icon (pencil).
- Field: Locate Rule Position (or Ruler position in your screenshot).
- Value: Change the current position from whatever it is now (likely "Below" or a low number) to Top (or Above Rule 1 if you have auto-added rules).
- Click Save and try rerunning the test.



- **"No matching rule (ID: 0)"** followed by the **Result: Blocked** is the Sophos Firewall's way of saying the traffic was blocked by the **Implicit Deny Rule**. (SUCCESS)

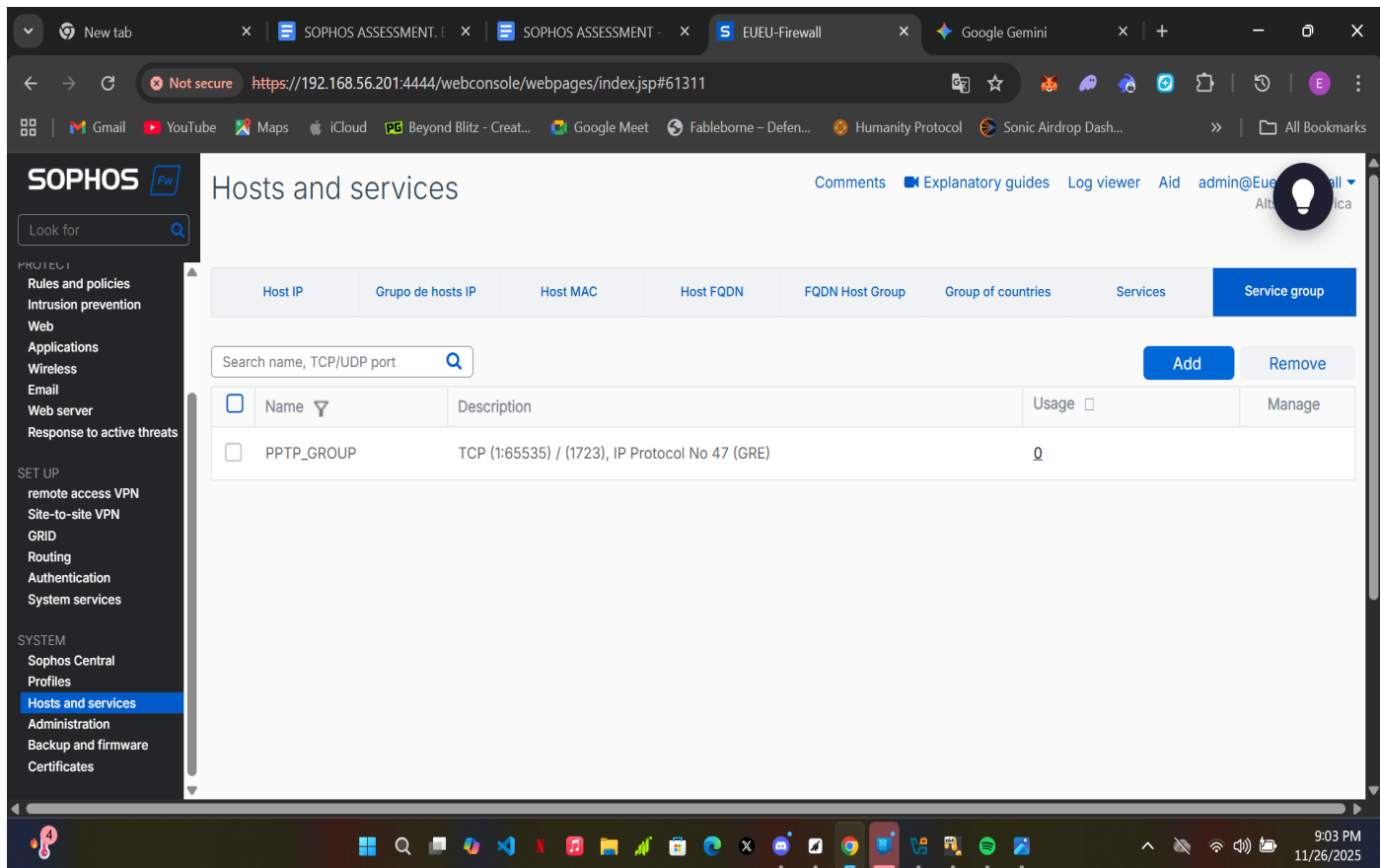
## POLICY NUMBER 2

### Blocking Outbound SSH/Telnet for the LAN

**Goal:** Prevent any internal host from making remote administrative connections outside the network.

#### Step 1: Create Service Group

- Hosts and services → Service Group → Add



- Description: Group for outbound blocking of SSH (22) and Telnet (23) access for the LAN zone.
- Select service Click Add new item → Select **SSH** (TCP 22) Adds the Secure Shell protocol and Click Add new item → Select **TELNET** (TCP 23)



- Save

SOPHOS FW

## Add service group

Comments Explanatory guides Log viewer Aid admin@Eue... All

Look for

PROJECT

- Rules and policies
- Intrusion prevention
- Web
- Applications
- Wireless
- Email
- Web server
- Response to active threats

SET UP

- remote access VPN
- Site-to-site VPN
- GRID
- Routing
- Authentication
- System services

SYSTEM

- Sophos Central
- Profiles
- Hosts and services**
- Administration
- Backup and firmware
- Certificates

| Host IP | Grupo de hosts IP | Host MAC | Host FQDN | FQDN Host Group | Group of countries | Services | Service group |
|---------|-------------------|----------|-----------|-----------------|--------------------|----------|---------------|
|---------|-------------------|----------|-----------|-----------------|--------------------|----------|---------------|

Name \*

SVC\_Remote\_Access\_Protocols

Description

Group for outbound blocking of SSH (22) and Telnet (23) access for the LAN zone.

Select service \*

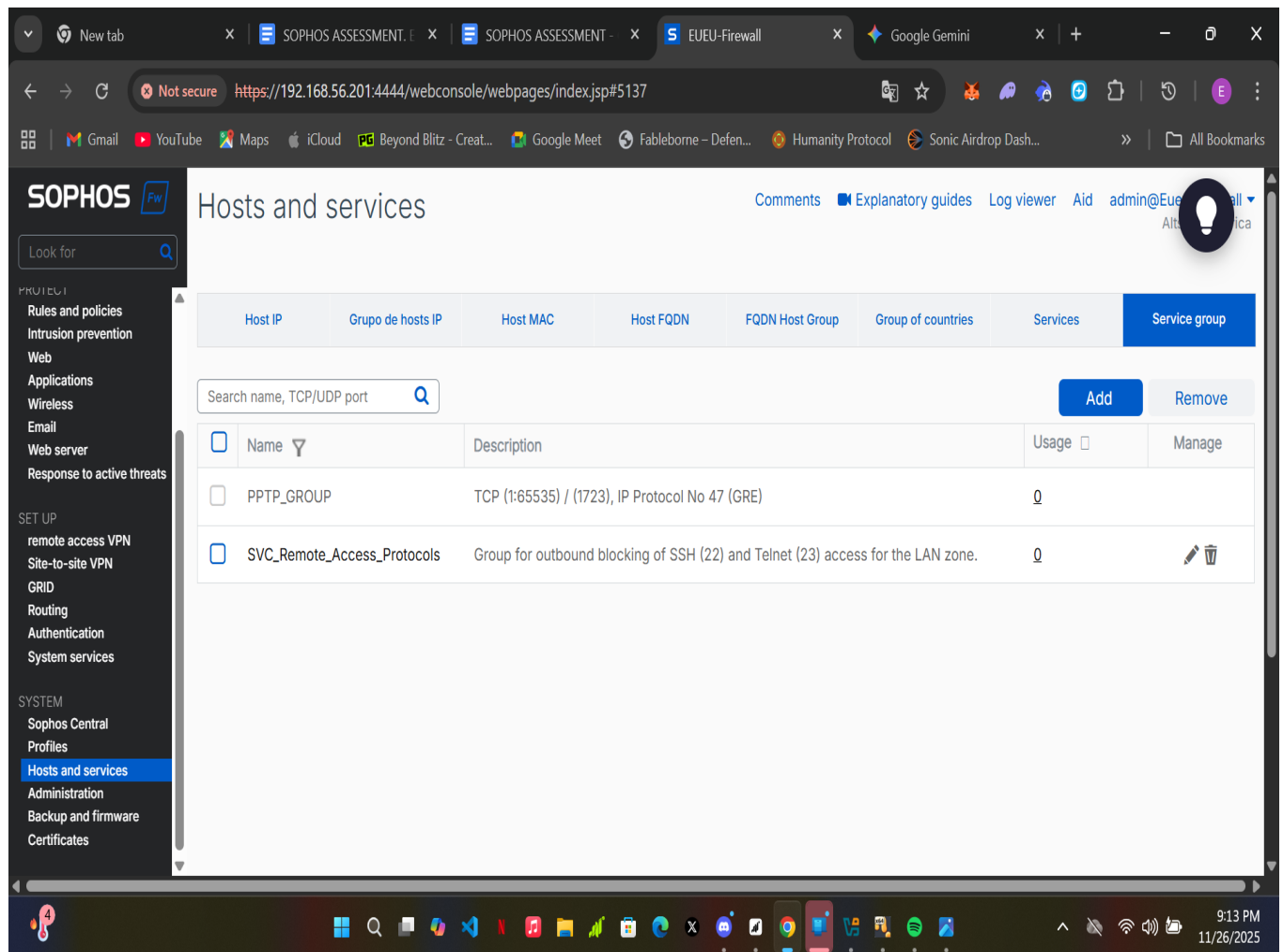
SSH

TELNET

Add new item

Save Cancel

9:12 PM 11/26/2025

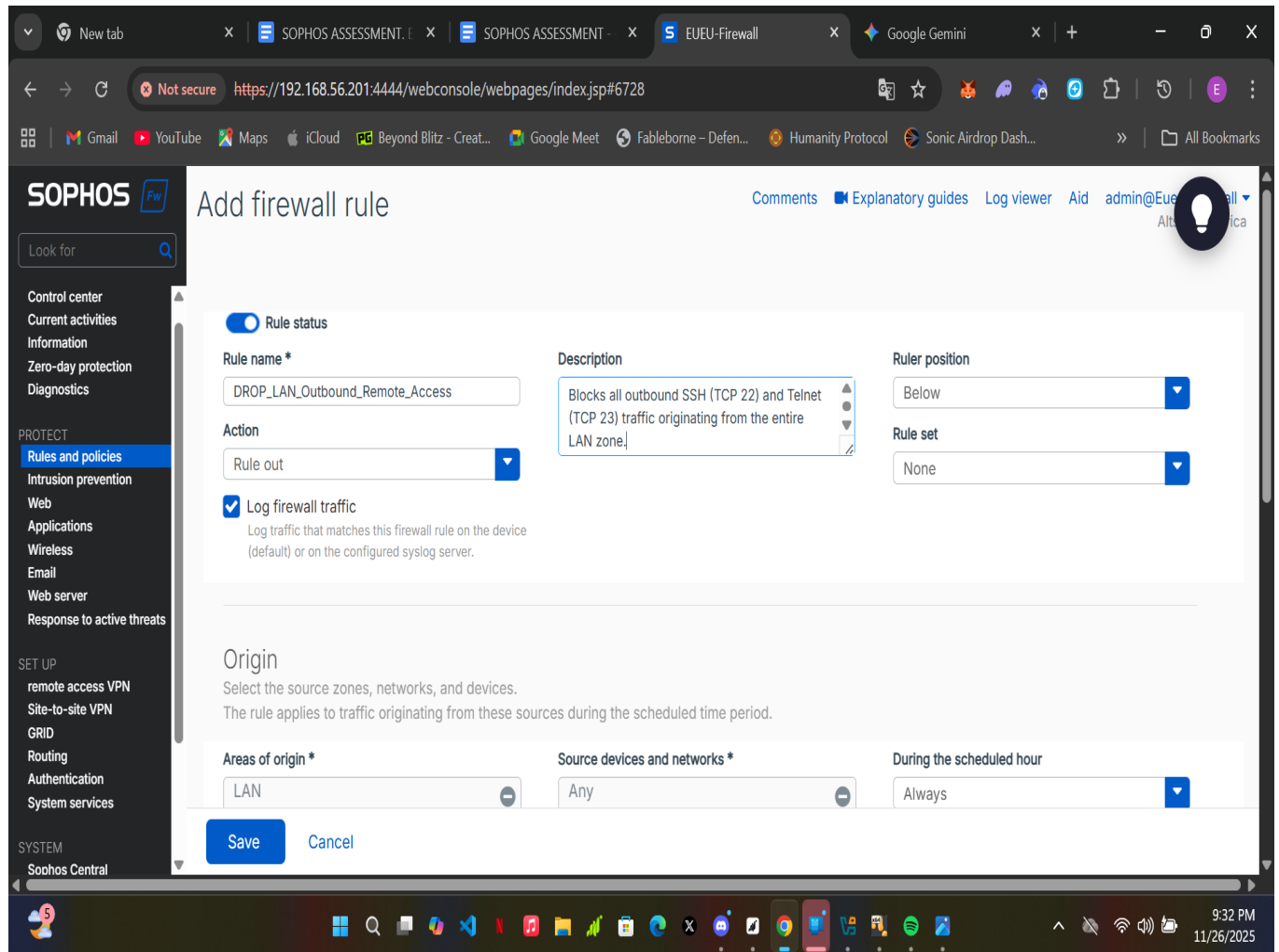


## Step 2: Create the Firewall Rule

- Navigation Path: Rules and Policies → Firewall Rules → Add Firewall Rule → New Firewall Rule.

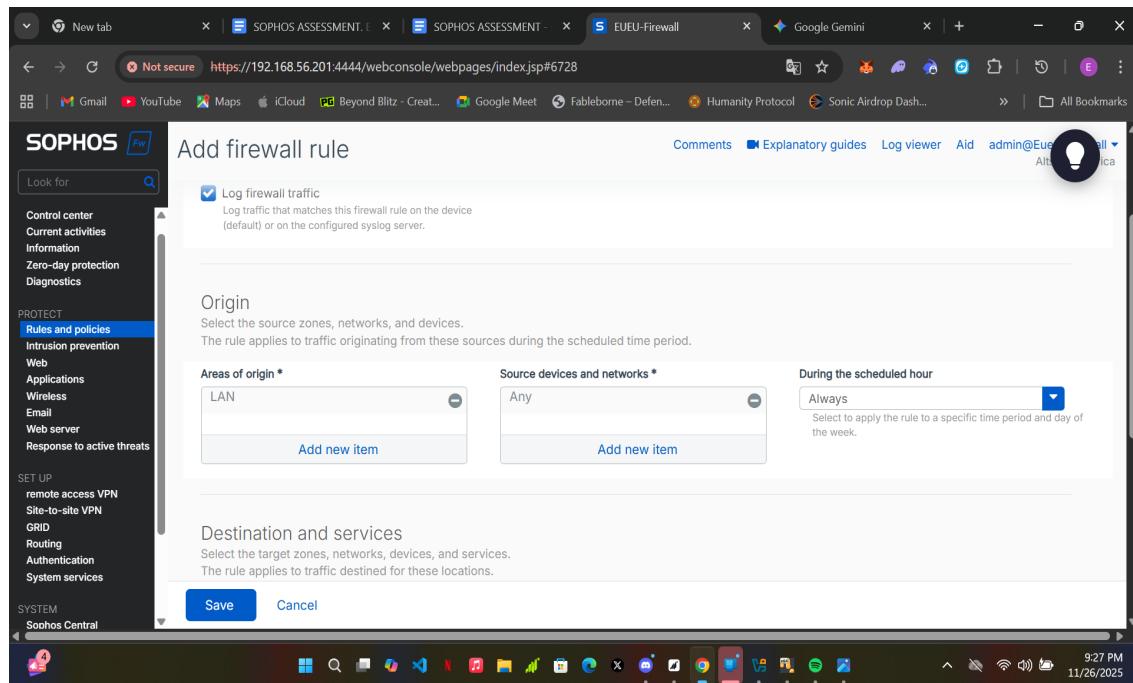
### General Settings:

- Rule Name: **DROP\_LAN\_Outbound\_Remote\_Access**
- Action: Rule out
- Blocks all outbound SSH (TCP 22) and Telnet (TCP 23) traffic originating from the entire LAN zone.
- Log firewall traffic: Check
- Rule set: None



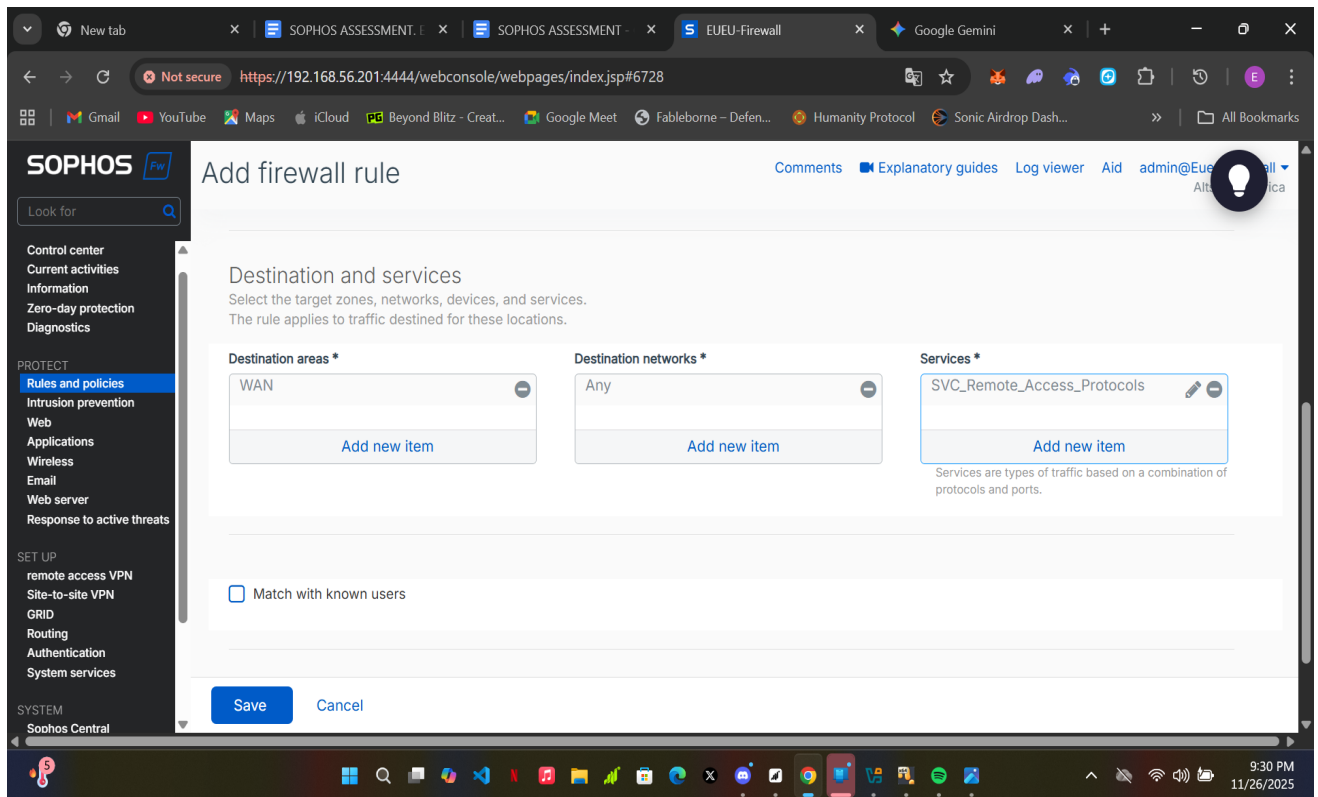
## Origin (Source):

- Areas of origin: Select LAN
- Source devices and networks: Select Any



## Destination and Services:

- Destination areas: Select WAN
- Destination networks: Select Any
- Services: Select **SVC\_Remote\_Access\_Protocols**
- Click save



- Analyze the Policy Test Results and confirm the rule set is working



